The EPCOS SIOV-B80K680 Block Varistor: Operating Characteristics and Application Sizing

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Abstract

This note describes the operating characteristics of the EPCOS SIOV-B80K680 metal-oxide block varistor (HighE series) starting from the manufacturer datasheet, derives an analytical model of its V–I characteristic, and develops a complete sizing example: line-side overvoltage protection of a high-power thyristor converter fed from a 600 V industrial network. The sizing covers the continuous operating voltage, the protection level and its coordination with the semiconductor blocking voltage, the single-pulse energy capability, and the average-power (repetition-rate) limit. A custom Simscape implementation of the device model is described and verified against the datasheet points.

1 Introduction

A metal-oxide varistor (MOV) is a voltage-dependent resistor based on sintered zinc-oxide (ZnO) ceramic. The ZnO grains are good conductors, while the grain boundaries behave as back-to-back Schottky-like barriers with a breakdown voltage of roughly 3.5 V each. A varistor disk is therefore electrically equivalent to a large series–parallel network of microvaristors: the disk thickness sets the varistor voltage, the disk area sets the current and energy capability. This is why, within the same voltage class, the EPCOS HighE series scales from the B32 (32 mm disk) up to the B80 (80 mm disk) with surge current ratings from 25 kA to 100 kA [1, 2].

The resulting terminal behavior is a strongly nonlinear, symmetric V–I characteristic, commonly approximated by the power law

$$i = k\, v^{\alpha}, \tag{1}$$

with a nonlinearity exponent α in the range 25–40 for ZnO. Three regions can be identified:

1. *Leakage region* (below the varistor voltage): the device is essentially a capacitor with a small, temperature-dependent resistive leakage current;
2. *Normal operation (clamping) region*: the grain-boundary barriers conduct and the power law (1) holds; the voltage rises only weakly with current;
3. *Upturn region* (tens of kA): the bulk resistance of the ZnO grains dominates and the characteristic becomes nearly ohmic.

The B80K680 is the 80 mm disk of the 680 V (AC) voltage class, i.e. the largest energy-handling member of its class in the HighE family, intended for heavy-duty protection of low-voltage industrial equipment: converter inputs, large drives, distribution panels and machine-tool supplies.

Table 1: SIOV-B80K680 ratings ($T_A = 85\,^\circ\mathrm{C}$) and characteristics ($T_A = 25\,^\circ\mathrm{C}$).

Quantity	Symbol	Value
Max. continuous AC voltage	V_{RMS}	680 V
Max. continuous DC voltage	V_{DC}	895 V
Max. surge current (8/20 µs, single pulse)	$i_{\max}$	100 kA
Max. single-pulse energy (2 ms)	$W_{\max}$	3600 J
Max. average power dissipation	$P_{\max}$	2.0 W
Varistor voltage at 1 mA (DC)	V_v	1100 V ± 10 %
Max. clamping voltage at 800 A (8/20 µs)	v_c	1815 V
Typical capacitance (1 kHz)	C_{typ}	5200 pF
Response time		< 25 ns
Operating temperature		−40 °C to 85 °C
Electric strength (housing)		≥ 2.5 kV RMS
Terminals / max. torque		M5 / 2.5 N m
Weight		approx. 650 g

2 Datasheet Data

Table 1 collects the ratings and characteristics of the B80K680 [1].

Two remarks on these figures:

- The energy rating $W_{\max}$ refers to a *single* 2 ms rectangular pulse. Repetitive surges must be checked against the derating curves of the datasheet, which trade pulse amplitude and duration against the admissible number of pulses over the device lifetime.
- $P_{\max}$ is a thermal steady-state limit: it bounds the *average* dissipation (standby losses plus the time-averaged surge energy) so that the disk temperature, and hence the leakage current, remains stable. ZnO leakage has a negative temperature coefficient, so exceeding $P_{\max}$ can trigger thermal runaway.

3 Analytical V–I Model

The datasheet provides two points of the characteristic: the varistor voltage (1 mA, 1100 V) and the clamping point (800 A, 1815 V). Fitting the power law (1) through both points gives the nonlinearity exponent

$$\alpha = \frac{\ln(i_2/i_1)}{\ln(v_2/v_1)} = \frac{\ln\!\left(800/10^{-3}\right)}{\ln(1815/1100)} = \frac{13.59}{0.501} \approx 27.1, \tag{2}$$

a typical value for ZnO. Inverting (1), the clamping voltage as a function of current is

$$v = C\, i^{\beta}, \qquad \beta = \frac{1}{\alpha} \approx 0.0369, \qquad C \approx 1419\,\mathrm{V}\ (\text{at } i = 1\,\mathrm{A}). \tag{3}$$

Equation (3) reproduces both datasheet points within 0.2 % and is plotted in Fig. 1. It is valid between the two fitting points; extrapolation towards the surge-current range (above a few kA) underestimates the upturn caused by the grain bulk resistance and should be replaced by the datasheet V/I curves for accurate coordination studies.

3.1 Capacitance and standby behavior

With $C_{\mathrm{typ}} = 5200\,\mathrm{pF}$, the capacitive reactance at 50 Hz is

$$X_C = \frac{1}{2\pi \cdot 50 \cdot 5.2 \times 10^{-9}} \approx 612\,\mathrm{k\Omega}, \tag{4}$$

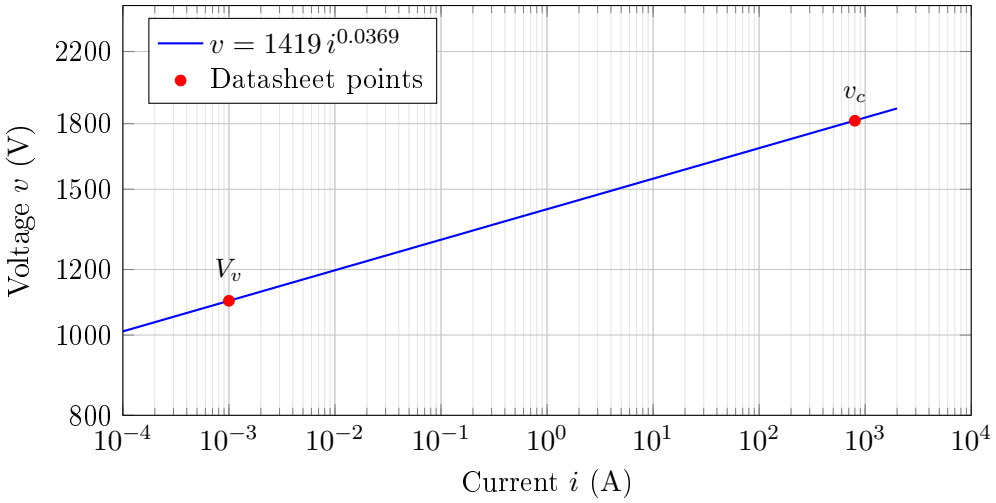


Figure 1: Fitted V–I characteristic of the B80K680 (clamping region).

so at the full rated voltage of 680 V RMS the capacitive current is about 1.1 mA — comparable to the 1 mA resistive reference current that defines V_v. At mains frequency the standby current of a large-disk varistor is therefore mostly capacitive and the standby losses remain far below $P_{\max}$. The same capacitance must be kept in mind on converter terminals with high $\mathrm{d}v/\mathrm{d}t$: 5.2 nF across a switching node is a non-negligible snubber-like load, which is one reason why block varistors are normally installed on the line side rather than directly across fast-switching devices.

4 Application Example: Line-Side Protection of a Thyristor Converter

4.1 System description

Consider a high-power thyristor converter (soft starter or controlled rectifier, hundreds of kW) fed from a 3×600 V, 50 Hz industrial network with $\pm 10\,\%$ voltage tolerance. The thyristors have a repetitive blocking voltage $V_{\mathrm{DRM}} = V_{\mathrm{RRM}} = 2200$ V. The installation is exposed to switching surges (breaker operation on transformers and reactors) and to induced lightning surges, assumed up to 10 kA with 8/20 µs waveshape. Three varistors are delta-connected at the converter input terminals (Fig. 2).

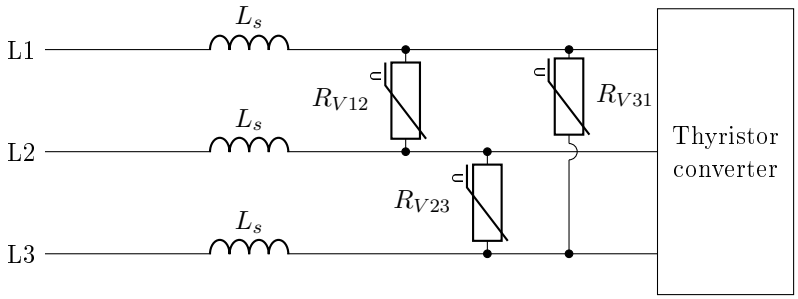


Figure 2: Delta-connected B80K680 varistors at the input of a thyristor converter. L_s is the source (line/transformer) inductance.

4.2 Step 1: continuous operating voltage

For line-to-line connection the varistor sees, in the worst case,

$$V_{\mathrm{LL,max}} = 1.1 \cdot 600\,\mathrm{V} = 660\,\mathrm{V} \;\le\; V_{\mathrm{RMS}} = 680\,\mathrm{V} \quad \checkmark \tag{5}$$

A useful additional check is that the network crest voltage never enters the conduction region even for the worst-case tolerance of the varistor voltage ($-10\,\%$, i.e. $V_{v,\min} = 990\,\mathrm{V}$):

$$\hat{v} = \sqrt{2} \cdot 660\,\mathrm{V} = 933\,\mathrm{V} \;<\; V_{v,\min} = 990\,\mathrm{V} \quad \checkmark \tag{6}$$

so the resistive standby current stays below the 1 mA reference level at every point of the mains cycle. Note that the 680 V class is the correct choice for a 600 V network; on a 690 V network ($+10\,\% \to 759\,\mathrm{V}$) the next class up (K750) would be required.

4.3 Step 2: protection level and coordination

At the tabulated clamping point the varistor limits the terminal voltage to $v_c = 1815\,\mathrm{V}$ at 800 A. The coordination margin with respect to the thyristor blocking voltage is

$$\frac{V_{\mathrm{DRM}}}{v_c} = \frac{2200}{1815} \approx 1.21, \tag{7}$$

i.e. about $20\,\%$, which is the usual minimum for insulation coordination. Two practical caveats:

- For the assumed 10 kA surge, the model (3) extrapolates to $v \approx 2.0\,\mathrm{kV}$, and the real curve (upturn region) is somewhat higher. The residual margin at full surge current is therefore thin: the datasheet V/I curve (page 285 of [1]) must be used for the final check, and this is precisely the reason to select the largest disk (B80) of the voltage class — for a given current, a larger disk clamps at a lower voltage.
- The connection inductance adds $L\,\mathrm{d}i/\mathrm{d}t$ to the clamping voltage seen by the converter. With the 8/20 µs front ($\mathrm{d}i/\mathrm{d}t \approx 1.25\,\mathrm{kA/\mu s}$ at 10 kA) each meter of loop ($\approx 1\,\mu\mathrm{H}$) adds about 1.25 kV. Varistor leads must be as short as possible and the varistor connected directly at the terminals to be protected.

4.4 Step 3: surge current and energy

Lightning-type surge. For a 10 kA, 8/20 µs pulse the absorbed energy can be bounded conservatively with an equivalent duration of 20 µs:

$$W_{\mathrm{surge}} \approx v_c\, \hat{\imath}\, t_{\mathrm{eq}} \approx 2000\,\mathrm{V} \cdot 10\,\mathrm{kA} \cdot 20\,\mu\mathrm{s} = 400\,\mathrm{J} \;\ll\; W_{\max} = 3600\,\mathrm{J} \quad \checkmark \tag{8}$$

The peak current itself is far below the single-pulse rating ($10\,\mathrm{kA} \ll 100\,\mathrm{kA}$); for the expected number of surges over the equipment lifetime the admissible amplitude is read from the derating curve (page 264 of [1]).

Inductive energy dump. The dimensioning event for the energy rating is usually not the lightning surge but the interruption of an inductive current: when the line breaker clears, the magnetic energy stored in the source and smoothing inductances is transferred to the varistors. Assuming the breaker interrupts $I_0 = 1.2\,\mathrm{kA}$ flowing in a total commutation/smoothing inductance $L = 1.5\,\mathrm{mH}$:

$$W_L = \tfrac{1}{2} L I_0^2 = \tfrac{1}{2} \cdot 1.5\,\mathrm{mH} \cdot (1.2\,\mathrm{kA})^2 = 1080\,\mathrm{J} \;=\; 0.30\, W_{\max} \quad \checkmark \tag{9}$$

During the dump the varistor clamps at $v \approx 1419 \cdot 1200^{0.0369} \approx 1.84\,\mathrm{kV}$ (from (3)), still coordinated with the thyristors, and the current decays approximately linearly in $t \approx L I_0 / v_c \approx 1\,\mathrm{ms}$, consistent with the 2 ms basis of the energy rating.

4.5 Step 4: average power and repetition rate

After a full 1080 J event the device needs a cool-down interval of at least

$$t_{\mathrm{rep}} \geq \frac{W_L}{P_{\max}} = \frac{1080}{2.0} = 540\,\mathrm{s}, \tag{10}$$

i.e. roughly one energy dump every ten minutes in continuous service. If the protection concept requires more frequent operations (e.g. a load-commutation duty rather than an emergency clamp), the varistor is the wrong device and an active clamp or a braking chopper must absorb the energy, with the varistor kept as last-resort protection only.

4.6 Result

A single B80K680 per phase pair (three devices, delta connection) satisfies all four checks for the 600 V converter: continuous voltage margin 3 % at +10 % grid tolerance, protection ratio 1.21 at the tabulated clamping point, single-pulse energy utilization 30 %, and a repetition capability of one full dump per 540 s. The tight coordination margin at 10 kA is the critical item and must be verified on the datasheet V/I curve; if the margin proves insufficient, the options are a higher thyristor blocking class or a hybrid scheme (varistor plus RC snubber to slow the front).

5 Practical Notes

- *End-of-life behavior.* A varistor degraded by repeated surges fails towards lower varistor voltage and eventually to a short circuit. Each device should be protected by a fuse or thermal disconnector able to clear the resulting fault current, and its state should be monitored (visual indicator or supervision contact) in permanently installed equipment.
- *Paralleling.* Because of the steep characteristic ($\alpha \approx 27$), small differences in V_v cause large current imbalance between paralleled varistors. Paralleling requires devices matched in V_v; it is generally preferable to select a single larger disk, which is exactly the role of the B80 size.
- *Mounting.* M5 screw terminals, max. torque 2.5 N m; the potted housing withstands $\geq$ 2.5 kV RMS to the mounting surface, but creepage and clearance of the external connections must still be verified for the 600 V network (overvoltage category and pollution degree of the installation).
- *Temperature.* All ratings of Table 1 hold up to $T_A = 85\,^{\circ}\mathrm{C}$; above that, voltage and energy must be derated per the datasheet derating curves.

6 Simscape Implementation

The device is implemented as a custom Simscape language component (`varistor_mov.ssc`) for the personal component library, so that it can be used in circuit simulations without requiring Simscape Electrical. Its default parameterization is the B80K680.

6.1 Equivalent circuit and equations

The model structure is shown in Fig. 3: a series inductance L_s and bulk resistance R_b feed the ZnO core, modeled as the parallel combination of the nonlinear resistor R_{nl}, the leakage resistance R_p and the capacitance C.

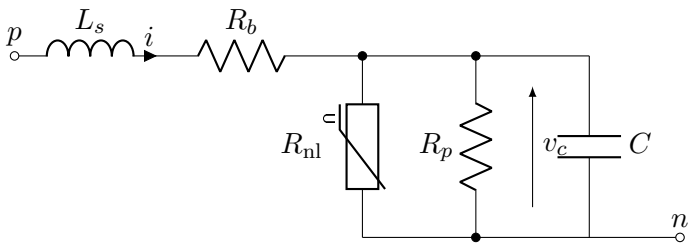


Figure 3: Structure of the Simscape varistor model.

The governing equations are

$$v = L_s \frac{\mathrm{d}i}{\mathrm{d}t} + R_b\, i + v_c, \tag{11}$$

$$i = C \frac{\mathrm{d}v_c}{\mathrm{d}t} + i_{\mathrm{nl}} + \frac{v_c}{R_p}, \tag{12}$$

$$i_{\mathrm{nl}} = I_1 \frac{v_c}{V_1} \left(\frac{|v_c|}{V_1} \right)^{\alpha - 1}, \qquad \alpha = \frac{\ln(I_2/I_1)}{\ln(V_2/V_1)}, \tag{13}$$

$$\frac{\mathrm{d}E_{\mathrm{abs}}}{\mathrm{d}t} = v_c \left(i_{\mathrm{nl}} + \frac{v_c}{R_p} \right) + R_b\, i^2. \tag{14}$$

The power law (13) is written in odd-symmetric form, $v_c\,|v_c|^{\alpha-1}$, which avoids the discontinuous sign$(\cdot)$ function and is differentiable at $v_c = 0$ — no zero-crossing events are generated. The component is parameterized directly with the two datasheet points (V_1, I_1) and (V_2, I_2), and α is computed internally per (13), exactly as in Section 3; no derived parameter has to be entered by hand. The three characteristic regions map onto the three core branches: R_p covers the leakage region (where the bare power law underestimates the real leakage current), the power law covers the clamping region (exact at the two fit points), and R_b approximates the upturn region.

Equation (14) integrates the *dissipated* energy only (core plus R_b, excluding the energy stored in C), so that E_{abs} is directly comparable with the single-pulse rating $W_{\max}$ of Table 1. An optional runtime assertion stops the simulation when $E_{\mathrm{abs}} > W_{\max}$; since the integral is cumulative, the check is meaningful for single-event simulations.

6.2 Parameters

Table 2 lists the model parameters with their B80K680 default values.

Table 2: Parameters of `varistor_mov` (defaults: B80K680).

Parameter	Default	Source	Role
V_1, I_1	1100 V, 1 mA	datasheet	varistor voltage point
V_2, I_2	1815 V, 800 A	datasheet	clamping point
C	5200 pF	datasheet	core capacitance
R_p	100 MΩ	assumed	leakage region
R_b	1 mΩ	placeholder	upturn region
L_s	50 nH	estimated	element + terminals
$W_{\max}$	3600 J	datasheet	energy assertion limit

Two parameters deserve comment. R_b is a placeholder: with 1 mΩ it adds only 100 V at 100 kA, so if surge currents above roughly 10 kA matter, R_b should be refitted on the upper part of the datasheet V/I curve. L_s must be kept strictly positive: it defines the current state of the

component (with $L_s = 0$ equation (11) degenerates into an algebraic constraint and the DAE index increases), and it represents in any case the physical inductance of the element and its terminals; external lead inductance is added in the surrounding circuit.

6.3 Verification and numerical remarks

Solving (11)–(13) at DC reproduces the datasheet points: $v(1\,\mathrm{mA}) = 1099.6\,\mathrm{V}$ and $v(800\,\mathrm{A}) = 1815.8\,\mathrm{V}$, where the residuals of about 1 V are the R_b and R_p contributions. At 10 kA the model gives $v \approx 2.0\,\mathrm{kV}$, consistent with the extrapolation discussed in Section 4.2 (and subject to the same caveat on the upturn region). At the network crest voltage of the sizing example (933 V) the resistive standby current is 21 µA, dominated by the R_p branch, confirming that at mains frequency the terminal current is essentially capacitive.

With $\alpha \approx 27$ the model is numerically stiff: the incremental resistance of the core spans more than eight decades between standby and clamping. A stiff solver (`ode23t` or `ode15s`) should be used, as for the other nonlinear device models of the library.

The B80K680 combines a 680 V continuous AC rating with 100 kA surge capability and 3600 J single-pulse energy, making it suitable as heavy-duty line-side protection for 600 V-class power converters. A two-point power-law fit of the datasheet characteristic ($\alpha \approx 27$, $v \approx 1419\, i^{0.0369}$) is sufficient for first-pass sizing of the clamping voltage and of the energy absorbed in inductive dump events, while the final coordination at full surge current must rely on the datasheet V/I and derating curves. In the worked example the device passes all sizing checks with the protection-level coordination as the binding constraint — the typical situation when protecting 2200 V thyristors on a 600 V network. The same two-point parameterization drives the custom Simscape component of Section 6, which reproduces the datasheet points and makes the device directly usable in transient circuit simulations.

References

[1] EPCOS AG, “Block Varistors, HighE Series,” SIOV Metal Oxide Varistors data book, pp. 221–226, Apr. 2002.

[2] EPCOS AG, “SIOV Metal Oxide Varistors — General Technical Information,” SIOV data book, 2002.

[3] J. D. Harnden, F. D. Martzloff, W. G. Morris, and F. B. Golden, “Metal-oxide varistor: a new way to suppress transients,” *Electronics*, vol. 45, no. 21, pp. 91–95, Oct. 1972.

[4] IEC 61643-11, *Low-voltage surge protective devices — Part 11: Surge protective devices connected to low-voltage power systems — Requirements and test methods*, International Electrotechnical Commission.